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Panel speaker

Abstract

A panel speaker includes: a single vibration panel formed by adhering one surface of a first panel and one surface of a second panel; a first vibration actuator configured to drive and vibrate the single vibration panel; and a second vibration actuator configured to drive and vibrate the single vibration panel, the single vibration panel has, in a plan view, a first non-adhesive region and a second non-adhesive region in both of which the first panel and the second panel are not bonded to each other, and the first vibration actuator is coupled to other surface of the second panel to drive the first non-adhesive region, and the second vibration actuator is coupled to the other surface of the second panel to drive the second non-adhesive region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-169247 filed on Oct. 21, 2022.

TECHNICAL FIELD

(2) The present invention relates to a panel speaker.

BACKGROUND ART

(3) There has been a display speaker that turns a display into a speaker by vibrating a display panel of the display using an actuator (vibration actuator) having a piezoelectric element. In a display speaker, a vibration actuator is attached to the back surface of the display panel (vibration panel). Thus, the vibration of the vibration actuator can be transmitted to the display panel. The display speaker is an example of panel speaker.

(4) As conventional arts, JP2019-191587A and JP2018-093468A are known.

(5) In the display panel, a plurality of (for example, two) independently controlled vibration

actuators are attached to the display panel to obtain a stereophonic display panel that outputs stereophonic sound. In some cases, a transmission panel is bonded to the back side of the display panel. Details of the transmission panel will be described later. The transmission panel is fixed to the display panel by a double-sided adhesive tape or the like. The vibration actuator is attached to the transmission panel and integrally drives the transmission panel and the display panel. That is, the transmission panel and the display panel are integrated to form the vibration panel. In a display panel attached with two vibration actuators, when one vibration actuator is vibrated, not only the periphery of the vibration actuator of the corresponding vibration panel but also the periphery of the other vibration actuator of the vibration panel may be vibrated. Moreover, in the display panel, when the other vibration actuator is vibrated, not only the periphery of the vibration actuator of the corresponding vibration panel but also the periphery of the one vibration actuator of the vibration panel may be vibrated. Therefore, in this case, the sound output from the display panel may be monaural sound instead of stereophonic sound. That is, the left/right separation (left/right average vibration ratio) in the display panel under stereophonic driving is low.
SUMMARY OF INVENTION
(6) An object of the technique of the present disclosure is to provide a panel speaker that improves the quality of output sound (in particular, the separation between left and right channels of stereophonic sound).
(7) As aspects of the invention, the following means are adopted.
(8) According to the first aspect, there is provided a panel speaker including: a single vibration panel formed by adhering one surface of a first panel and one surface of a second panel; a first vibration actuator configured to drive and vibrate the single vibration panel; and a second vibration actuator configured to drive and vibrate the single vibration panel, wherein the single vibration panel has, in a plan view, a first non-adhesive region and a second non-adhesive region in both of which the first panel and the second panel are not bonded to each other, and the first vibration actuator is coupled to other surface of the second panel to drive the first non-adhesive region of the single vibration panel, and the second vibration actuator is coupled to the other surface of the second panel to drive the second non-adhesive region of the single vibration panel.
(9) According to the disclosed technique, it is possible to provide a panel speaker that improves the quality of output sound (in particular, the separation between left and right channels of stereophonic sound).

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is an exploded perspective view of a display speaker **800**.
(2) FIG. **2** is a cross-sectional view of a plane parallel to an xz plane passing through a first vibration actuator **851** and a second vibration actuator **852** of the display speaker **800**.
(3) FIG. **3** is a back view of a transmission panel **830**, an adhesive layer **820**, a first coupling portion **841**, and a second coupling portion **842**.
(4) FIG. **4** is an exploded perspective view of a display speaker **200**.
(5) FIG. **5** is a cross-sectional view of a plane parallel to the xz plane passing through a first vibration actuator **251** and a second vibration actuator **252** of the display speaker **200**.
(6) FIG. **6** is a back view of a transmission panel **230**, an adhesive layer **220**, a first coupling portion **241**, and a second coupling portion **242**.
(7) FIG. **7** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to a first modification.
(8) FIG. **8** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling

portion **241**, and the second coupling portion **242** of the display speaker **200** according to a second modification.

(9) FIG. **9** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to a third modification.

(10) FIG. **10** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to a fourth modification.

(11) FIG. **11** is a diagram (**1**) illustrating simulation results of vibration of the transmission panel **230** (display panel **210**) of the display speaker **200** of the present embodiment.

(12) FIG. **12** is a diagram (**2**) illustrating simulation results of vibration of the transmission panel **230** (display panel **210**) of the display speaker **200** of the present embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(13) Hereinafter, embodiments will be described with reference to the drawings. A configuration of the embodiments is illustrative, and the configuration of the invention is not limited to a specific configuration of the embodiments of the disclosure. In accordance with the embodiments of the invention, the specific configuration according to the embodiments may be adopted as appropriate.

(14) Here, a case where a plurality of vibration actuators are attached to a transmission panel, the transmission panel is provided on a display panel of a display, and the display is caused to function as a display speaker under stereophonic driving will be described. A panel speaker is a device that outputs (emits) sound waves (sound) from the display panel by vibrating the display panel with the vibration actuators. The display speaker is an example of the panel speaker. The panel speaker is, for example, a speaker that outputs sound by vibrating a planar panel (vibration panel) such as a display panel.

EMBODIMENTS

Comparative Example

(15) FIGS. **1**,**2**, and **3** are diagrams illustrating a configuration example of a display speaker **800** according to a comparative example of the present embodiment. FIG. **1** is an exploded perspective view of the display speaker **800**. The display speaker **800** includes a display panel **810**, an adhesive layer **820**, a transmission panel **830**, a first coupling portion **841**, a second coupling portion **842**, a first vibration actuator **851**, and a second vibration actuator **852**. Here, in a plan view of the display panel **810** from the front side, a direction from left to right is defined as an x direction, a direction from bottom to top is defined as a y direction, and a direction from front to back is defined as a z direction. The x direction, the y direction, and the z direction are orthogonal to one another. FIG. **2** is a cross-sectional view of a plane parallel to an xz plane passing through the first vibration actuator **851** and the second vibration actuator **852** of the display speaker **800**. FIG. **3** is a back view of the transmission panel **830**, the adhesive layer **820**, the first coupling portion **841**, and the second coupling portion **842**.

(16) The display panel **810** is a plate-shaped rectangular panel. Each side of the outer shape of the display panel **810** is parallel to the x-axis or the y-axis.

(17) The adhesive layer **820** joins the back side of the display panel **910** and the front side of the transmission panel **930**. In the example of FIG. **1**, the adhesive layer **820** has substantially the same shape and size as the display panel **810** and the transmission panel **830**. The adhesive layer **820** is, for example, a double-sided adhesive tape. In the example of FIG. **1**, the adhesive layer **820** is rectangular.

(18) The transmission panel **830** is disposed on the back side of the display panel **810**, between the display panel **810** and the first vibration actuator **851**, and between the display panel **810** and the second vibration actuator **852**. The transmission panel **830** is a rectangular panel having substantially the same size as the display panel **810** when viewed from the front. The transmission panel **830** is, for example, a metal plate such as an aluminum plate. The transmission panel **830** has

functions of supporting and reinforcing the display panel **810** from the back side, dissipating heat generated in the display panel **810**, shielding electromagnetic noise generated in the display panel **810**, and the like. The transmission panel **830** does not necessarily have all of the above-described roles, and may be, for example, any insulator such as a glass plate or a plastic plate instead of a metal plate as long as intended for only reinforcement. The transmission panel **830** transmits the vibrations of the first vibration actuator **851** and the second vibration actuator **852** to the display panel **810**. In other words, the transmission panel **830** and the display panel **810** are integrally driven by the first vibration actuator **851** and the second vibration actuator **852**, and constitute a vibration panel that outputs sound waves from the surface closer to the display panel **810**. The functions of the transmission panel **830** and the display panel **810** are common to the transmission panel and the display panel of all the embodiments described below.

(19) The first coupling portion **841** is, for example, disposed over the entire front surface side of the first vibration actuator **851**, and couples the back surface of the transmission panel **830** and the front surface of the first vibration actuator **851**. The first coupling portion **841** is, for example, a double-sided adhesive tape. The first coupling portion **841** is disposed at a position translated in the −x direction from the center of the transmission panel **830** in a plan view. The second coupling portion **842** is, for example, disposed over the entire front surface side of the second vibration actuator **852**, and couples the back surface of the transmission panel **830** and the front surface of the second vibration actuator **852**. The second coupling portion **842** is, for example, a double-sided adhesive tape. The second coupling portion **842** is disposed at a position translated in the +x direction from the center of the transmission panel **830** in a plan view. The first coupling portion **841** and the second coupling portion **842** are equal in distance from the center of the display panel **810** (transmission panel **830**) in the x direction. The first coupling portion **841** and the second coupling portion **842** are located at positions in line symmetry with respect to a straight line passing through the center of the display panel **810** and parallel to the y-axis as a center axis of symmetry.

(20) The first vibration actuator **851** and the second vibration actuator **852** are disposed on the back surface side of the transmission panel **830**. The first vibration actuator **851** and the second vibration actuator **852** are plate-shaped rectangular members smaller than the transmission panel **830**. The first vibration actuator **851** and the second vibration actuator **852** are, for example, piezoelectric elements. The first vibration actuator **851** and the second vibration actuator **852** expand and contract according to an applied driving voltage to vibrate. The first vibration actuator **851** and the second vibration actuator **852** are controlled independently of each other.

(21) In the display speaker **800**, when the first vibration actuator **851** vibrates, not only the periphery of the first vibration actuator **851** of the transmission panel **830** but also the periphery of the second vibration actuator **852** of the transmission panel **830** vibrates. In the display speaker **800**, when the second vibration actuator **852** vibrates, not only the periphery of the second vibration actuator **852** of the transmission panel **830** but also the periphery of the first vibration actuator **851** of the transmission panel **830** vibrates. Thus, even if separate signals are input to the first vibration actuator **851** and the second vibration actuator **852** to vibrate them in order to output stereophonic sound, the display speaker **800** outputs sound such as monaural sound. That is, in the display speaker **800**, the left/right separation of sound (left/right average vibration ratio) is low. At this time, the left/right average vibration ratio is 1.1. The left/right average vibration ratio is, for example, a ratio (PL/PR) of an average vibration magnitude (amplitude or acceleration) PL in a region on the left side of the transmission panel **830** (in the vicinity of the first vibration actuator **851** of the transmission panel **830**) to an average vibration magnitude (amplitude or acceleration) PR in a region on the right side of the transmission panel **830** (in the vicinity of the second vibration actuator **852** of the transmission panel **830**) when the left vibration actuator (the first vibration actuator **851**) is vibrated and the right vibration actuator (the second vibration actuator **852**) is not vibrated. Here, the “average magnitude of vibration in a region” can be defined as, for

example, a value obtained by integrating the magnitude of vibration in micro regions included in the region over the entire region, or a value obtained by dividing the value by the area of the region. In a simplified manner, it can be defined as the sum of the magnitudes of vibration at several measurement points included in the region, or a value obtained by dividing the sum by the number of the measurement points. When the left/right average vibration ratio (PL/PR) is 1 or more, the left/right separation of sound increases as the left/right average vibration ratio is larger, and decreases as the left/right average vibration ratio is closer to 1. When the right vibration actuator (second vibration actuator **852**) is vibrated and the left vibration actuator (first vibration actuator **851**) is not vibrated, the left/right average vibration ratio can be defined by PR/PL.

Configuration Example

(22) FIGS. **4**,**5**, and **6** are diagrams illustrating configuration examples of the display speaker **200** of the present embodiment. FIG. **4** is an exploded perspective view of the display speaker **200**. The display speaker **200** includes a display panel **210**, an adhesive layer **220**, a transmission panel **230**, a first coupling portion **241**, a second coupling portion **242**, a first vibration actuator **251**, and a second vibration actuator **252**. Here, in a plan view of the display panel **210** from the front side, a direction from left to right is defined as the x direction, a direction from bottom to top is defined as the y direction, and a direction from front to back is defined as the z direction. The x direction, the y direction, and the z direction are orthogonal to one another. FIG. **5** is a cross-sectional view of a plane parallel to the xz plane passing through the first vibration actuator **251** and the second vibration actuator **252** of the display speaker **200**. FIG. **6** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242**. Here, the x direction is the longitudinal direction of the display panel **210**, and they direction is the lateral direction of the display panel **210**.

(23) The display panel **210** is a plate-shaped rectangular panel. Each side of the outer shape of the display panel **210** is parallel to the x-axis or the y-axis. That is, two opposite sides of the rectangle of the outer shape of the display panel **210** are parallel to a first direction (x direction), and the other two opposite sides are parallel to a second direction (y direction) orthogonal to the first direction. The display panel **210** is, for example, a planar glass panel or an organic light emitting diode (OLED) panel. The display panel **210** is joined to the transmission panel **230** via the adhesive layer **220**. The display panel **210** is an example of a first panel. The shape of the display panel **210** and the like is not limited to a rectangular shape, and may be other shapes such as a quadrangle shape other than rectangle, a triangular shape, a polygonal shape, an elliptical shape, and a circular shape. The display panel **210** is, for example, a display panel for displaying character information and images by a computer or the like.

(24) The adhesive layer **220** is divided into a plurality of regions, and joins the back side of the display panel **210** and the front side of the transmission panel **230**. In the example of FIG. **4**, the adhesive layer **220** includes a first adhesive region **221**, a second adhesive region **222**, and a third adhesive region **223**. Each adhesive region of the adhesive layer **220** is, for example, a double-sided adhesive tape. Each adhesive region of the adhesive layer **220** may be an adhesive. The adhesive regions are arranged at different positions so as not to overlap each other in a plan view (when viewed from the front side or the back side, and when viewed from a direction orthogonal to the x direction and the y direction). The first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223** are strip-shaped rectangles parallel to the y-axis. The first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223** extend from the upper portion to the lower portion of the display panel **210** (transmission panel **230**). The first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223** are parallel to each other. The first adhesive region **221** and the third adhesive region **223** have the same shape. The first adhesive region **221** and the second adhesive region **222** may have the same shape. Each of the first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223** has a width in the y direction larger than a width in the x direction. Here, the y direction

is the longitudinal direction of the first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223**. In the x direction, a distance (X**11**) between the first adhesive region **221** and the center of the display panel **210** is equal to a distance (X**13**) between the third adhesive region **223** and the center of the display panel **210**. In a plan view, the first adhesive region **121** and the third adhesive region **223** are located at positions in point symmetry with respect to the center of the display panel **210**. The shape of the second adhesive region **222** is in line symmetry with respect to a straight line passing through the center of the display panel **210** and parallel to the y-axis as the center axis of symmetry. The same applies to the relationship between the center of the transmission panel **130** and the adhesive regions. The display panel **210** and the transmission panel **230** are not joined in portions other than the respective adhesive regions of the adhesive layer **220** between the display panel **210** and the transmission panel **230**. That is, the portion other than the regions of the adhesive layer **220** between the display panel **210** and the transmission panel **230** is, for example, a gap (non-adhesive region). The portion other than the respective adhesive regions of the adhesive layer **220** between the display panel **210** and the transmission panel **230** is non-adhesive regions. Unlike the comparative example described above, in a plan view, the portion overlapping the first vibration actuator **251** (first coupling portion **241**) and the second vibration actuator **252** (second coupling portion **242**) does not include adhesive regions of the adhesive layer **220**. The number of adhesive regions of the adhesive layer **220** is not limited to three. The distance between any position and region (for example, the adhesive region) is the distance between the position and the center of the region. The center of the rectangle of the display panel **210** and the like is obtained as, for example, the intersection position of the diagonal lines of the rectangle.

(25) The center used here is not limited to the geometric center, and may be within a certain range including the geometric center. The certain range including the geometric center is, for example, a range of a circle having a radius of 2 cm around the geometric center, or a range of a circle having a radius of 5% of the maximum width of the display panel **210** around the geometric center. The center of the display panel **210** or the like may be defined by the center of gravity, the incenter, the circumcenter, the intersection of the center lines of the widths in two orthogonal directions, or the like. The first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223** may not be parallel to the outer shape of the display panel **210**. The shape of each adhesive region is not limited to that described herein. The adhesive regions are in line symmetry with respect to the straight line passing through the center of the display panel **210** and parallel to the y-axis. The adhesive layer **220** does not include a non-adhesive region. The adhesive layer **220** partitions the longitudinal direction of the display panel **210** (x direction) into a plurality of non-adhesive regions. In general, the longitudinal direction of the display panel **210** is the left-right direction of the display panel **210**. By providing the plurality of non-adhesive regions in this direction, the display speaker **200** can output sound from the left side based on the sound signal on the left (left sound signal) and output sound from the right side based on the sound signal on the right (right sound signal).

(26) The transmission panel **230** is disposed on the back side of the display panel **210**, between the display panel **210** and the first vibration actuator **251**, and between the display panel **210** and the second vibration actuator **252**. The transmission panel **230** is a rectangular panel having substantially the same size as the display panel **210** when viewed from the front (in a plan view). The transmission panel **230** transmits the vibrations of the first vibration actuator **251** and the second vibration actuator **252** to the display panel **210**. The transmission panel **230** is, for example, a metal plate such as an aluminum plate. In a plan view, the position of the center of the transmission panel **230** and the position of the center of the display panel **210** are the same. The transmission panel **230** is an example of a second panel.

(27) The first coupling portion **241** is, for example, disposed over the entire front surface side of the first vibration actuator **251**, and couples the back surface of the transmission panel **230** and the front surface of the first vibration actuator **251**. The second coupling portion **242** is, for example,

disposed over the entire front surface side of the second vibration actuator **252**, and couples the back surface of the transmission panel **230** and the front surface of the second vibration actuator **252**. The first coupling portion **241** and the second coupling portion **242** are, for example, double-sided adhesive tapes. The first coupling portion **241** and the second coupling portion **242** may be an adhesive. The first coupling portion **241** and the second coupling portion **242** are examples of coupling portions. Instead of the first coupling portion **241** and the second coupling portion **242**, bosses pressed into the transmission panel **230** and screws fixing the first vibration actuator **251** and the second vibration actuator **252** to the boss may serve as coupling portions. At this time, screw holes for the screws to pass through are provided in the first vibration actuator **251** and the second vibration actuator **252**. The transmission panel **130**, the first vibration actuator **251**, and the second vibration actuator **252** are coupled by the coupling portions. The coupling portions that couple the transmission panel **130** to the first vibration actuator **251** and the second vibration actuator **252** are not limited to those described herein, and the coupling may be achieved by another configuration. The coupling portions are in line symmetry with respect to the straight line passing through the center of the display panel **210** and parallel to the y-axis.
(28) The first vibration actuator **251** and the second vibration actuator **252** are disposed on the back surface side of the transmission panel **230**. The first vibration actuator **251** and the second vibration actuator **252** are plate-shaped rectangular members smaller than the transmission panel **230**. The first vibration actuator **251** and the second vibration actuator **252** are, for example, piezoelectric elements. The first vibration actuator **251** and the second vibration actuator **252** expand and contract according to an applied driving voltage to vibrate. The first vibration actuator **251** and the second vibration actuator **252** are controlled independently of each other. The first vibration actuator **251** is disposed, for example, at a position translated in the −x direction from the center of the transmission panel **230** in a plan view. Here, the distance in the x direction between the center of the transmission panel **230** and the first vibration actuator **251** is X**21**. That is, the first vibration actuator **251** is disposed on a straight line that passes through the center of the transmission panel **250** and is parallel to the x direction. The second coupling portion **242** is disposed at a position translated in the +x direction from the center of the transmission panel **230** in a plan view. Here, the distance in the x direction between the center of the transmission panel **230** and the second vibration actuator **252** is X**22**. That is, the second vibration actuator **252** is disposed on a straight line that passes through the center of the transmission panel **250** and is parallel to the x direction. The first vibration actuator **251** and the second vibration actuator **252** are equal in distance from the center of the display panel **210** (transmission panel **230**) in the x direction (X**21**=X**22**). The first vibration actuator **251** and the second vibration actuator **252** are located at positions in line symmetry with respect to a straight line passing through the center of the display panel **210** and parallel to the y-axis as a center axis of symmetry. The first vibration actuator **251** and the second vibration actuator **252** are examples of a vibration unit. The first vibration actuator **251** and the second vibration actuator **252** may have a shape other than a rectangular shape. For example, the display speaker **200** can output stereophonic sound by inputting a left sound signal of stereophonic sound to the first vibration actuator **251** and inputting a right sound signal of stereophonic sound to the second vibration actuator **252**.
(29) In a plan view, the first vibration actuator **251** is disposed in the non-adhesive region between the first adhesive region **221** and the second adhesive region **222** (first non-adhesive region **261**). In a plan view, the second vibration actuator **252** is disposed in the non-adhesive region between the third adhesive region **223** and the second adhesive region **222** (second non-adhesive region **262**). In a plan view, the first non-adhesive region **261** and the second non-adhesive region **262** are different non-adhesive regions partitioned by the second adhesive region **222**. The first non-adhesive region **261** and the second non-adhesive region **262** are strip-shaped regions. Since in a plan view the first non-adhesive region **261** and the second non-adhesive region **262** are partitioned by the second adhesive region **222**, the vibration of the first vibration actuator **251** attached to the first non-

adhesive region **261** is less likely to be transmitted to the second non-adhesive region **262** of the transmission panel **230**. Similarly, the vibration of the second vibration actuator **252** attached to the second non-adhesive region **262** of the transmission panel **230** is less likely to be transmitted to the first non-adhesive region **261** of the transmission panel **230**. This facilitates separation between the vibration caused by the first vibration actuator **251** and the vibration caused by the second vibration actuator **252**. At this time, the left/right average vibration ratio of the display speaker **200** is 1.3. Accordingly, the left/right separation of the display speaker **200** is higher than that of the display speaker **800** of the comparative example. According to the display speaker **200**, the stereophonic performance is improved.

(30) Unlike the comparative example, in a plan view, the transmission panel **230** and the display panel **210** are not joined at positions where the first vibration actuator **251** and the second vibration actuator **252** are joined to the transmission panel **230**. Since in a plan view the transmission panel **230** and the display panel **210** are not joined at the positions where the first vibration actuator **251** and the second vibration actuator **252** are joined to the transmission panel **230**, resistance to vibration is prevented and the display panel **210** is easily vibrated. In a plan view, the transmission panel **230** and the display panel **210** are joined between the first vibration actuator **251** and the second vibration actuator **252**. Accordingly, the vibration of the first vibration actuator **251** and the vibration of the second vibration actuator **252** are less likely to be mixed on the transmission panel **230**.

(31) That is, in the display speaker **200** of the present embodiment, since the two non-adhesive regions are provided between the transmission panel **230** and the display panel **210** in a manner partitioned by the adhesive regions in a plan view, the quality of the sound output from the display speaker **200** (in particular, the separation of the left and right channels of the stereophonic sound) can be enhanced. In particular, in the related art, when a plurality of vibration actuators are coupled to one vibration panel to output stereophonic sound, a rib (wall) may be provided on the vibration panel to partition the vibration actuators so that vibrations by the left and right vibration actuators are not mixed on the vibration panel, but in the display speaker **200** of the present embodiment, it is not necessary to provide such a wall (rib), which simplifies the structure is simplified and is advantageous for cost reduction.

(32) FIGS. **11** and **12** are diagrams illustrating simulation results of vibration of the transmission panel **230** (display panel **210**) of the display speaker **200** of the present embodiment. FIG. **11** illustrates a graph of simulation results, and an area of the graph of FIG. **11** (parallel quadrilateral area) corresponds to the transmission panel **230** (or the display panel **210**) illustrated in FIG. **12**. The z-axis direction of the graph represents the amplitude or acceleration of vibration. FIG. **11** shows the vibration of the transmission panel **230** (or the display panel **210**) when the second vibration actuator **252** is vibrated and the first vibration actuator **251** is not vibrated. In FIG. **11**, there is a peak at a position corresponding to the second vibration actuator **252** in the graph, which indicates that a large amplitude or acceleration of the vibration, while there is no peak at a position corresponding to the first vibration actuator **251** in the graph, which indicates that a small amplitude or acceleration of the vibration. That is, it can be seen that the left/right average vibration ratio of the display speaker **200** is large.

(33) [First Modification]

(34) Next, a first modification will be described. The first modification is in common with the above-described configuration example. Accordingly, different points will be mainly described, and description of common points will be omitted. The same reference numerals are used for components having the same configurations as those of the above-described configuration example.

(35) FIG. **7** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to the first modification. Similar to the above configuration example, the display speaker **200** of the first modification includes the adhesive layer **220** including the first adhesive region **221**, the second

adhesive region **222**, and the third adhesive region **223**. The adhesive layer **220** includes an adhesive region **224-1** connecting the upper portion of the first adhesive region **221** and the upper portion of the second adhesive region **222**, and an adhesive region **224-2** connecting the lower portion of the first adhesive region **221** and the lower portion of the second adhesive region **222**. The adhesive layer **220** includes an adhesive region **224-3** connecting the upper portion of the third adhesive region **223** and the upper portion of the second adhesive region **222**, and an adhesive region **224-4** connecting the lower portion of the third adhesive region **223** and the lower portion of the second adhesive region **222**. The adhesive region **224-1**, the adhesive region **224-2**, the adhesive region **224-3**, and the adhesive region **224-4** are strip-shaped rectangles parallel to the x-axis. The first non-adhesive region **261** is a region surrounded by the first adhesive region **221**, the second adhesive region **222**, the adhesive region **224-1**, and the adhesive region **224-2**. The second non-adhesive region **262** is a region surrounded by the third adhesive region **223**, the second adhesive region **222**, the adhesive region **224-3**, and the adhesive region **224-4**. Surrounding the non-adhesive regions by the adhesive regions further facilitates separation between the vibration caused by the first vibration actuator **251** and the vibration caused by the second vibration actuator **252**. The left/right average vibration ratio of the display speaker **200** of the first modification is 1.6.
(36) [Second Modification]
(37) Next, a second modification will be described. The second modification is in common with the above-described configuration example. Accordingly, different points will be mainly described, and description of common points will be omitted. The same reference numerals are used for components having the same configurations as those of the above-described configuration example.
(38) FIG. **8** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to the second modification. Similar to the above configuration example, the display speaker **200** of the second modification includes the adhesive layer **220** including the first adhesive region **221** and the third adhesive region **223**. The adhesive layer **220** includes an adhesive region **225-1** and an adhesive region **225-2** instead of the second adhesive region **222**. The adhesive regions **225-1** and **225-2** are strip-shaped rectangles parallel to the y-axis. The adhesive regions **225-1** and **225-2** extend from the upper portion to the lower portion of the display panel **210** (transmission panel **230**). In the x direction, the distance between the adhesive region **225-1** and the center of the display panel **210** is equal to the distance between the adhesive region **225-2** and the center of the display panel **210**. The adhesive region **225-1** is disposed on the −x side from the center of the display panel **210**. The adhesive region **225-2** is disposed on the +x side from the center of the display panel **210**. A non-adhesive region **263** is provided between the adhesive region **225-1** and the adhesive region **225-2**. The first non-adhesive region **261** is a region surrounded by the first adhesive region **221** and the adhesive region **225-1**. The second non-adhesive region **262** is a region surrounded by the third adhesive region **223** and the adhesive region **225-2**. Separating the first non-adhesive region **261** and the second non-adhesive region **262** by the adhesive region **225-1**, the non-adhesive region **263**, and the adhesive region **225-2**, further facilitates separation between the vibration caused by the first vibration actuator **251** and the vibration caused by the second vibration actuator **252**.
(39) [Third Modification]
(40) Next, a third modification will be described. The third modification is in common with the above-described first modification. Accordingly, different points will be mainly described, and description of common points will be omitted. The same reference numerals are used for components having the same configurations as those of the above-described first modification.
(41) FIG. **9** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to the third modification. Similarly to the first modification, the display speaker **200** of the third modification includes the adhesive layer **220** including the first adhesive region **221**, the second adhesive region

222, the third adhesive region **223**, the adhesive region **224-1**, the adhesive region **224-2**, the adhesive region **224-3**, and the adhesive region **224-4**. In addition, the adhesive layer **220** includes an adhesive region **224-5** connecting the central portion of the first adhesive region **221** and the central portion of the second adhesive region **222**, and an adhesive region **224-6** connecting the central portion of the third adhesive region **223** and the central portion of the second adhesive region **222**. The adhesive regions **224-5** and **224-6** are strip-shaped rectangles parallel to the x-axis. The first non-adhesive region **261** is a region surrounded by the first adhesive region **221**, the second adhesive region **222**, the adhesive region **224-1**, and the adhesive region **224-5**. The second non-adhesive region **262** is a region surrounded by the third adhesive region **223**, the second adhesive region **222**, the adhesive region **224-3**, and the adhesive region **224-6**. The non-adhesive region **264** is a region surrounded by the first adhesive region **221**, the second adhesive region **222**, the adhesive region **224-2**, and the adhesive region **224-5**. The non-adhesive region **265** is a region surrounded by the third adhesive region **223**, the second adhesive region **222**, the adhesive region **224-4**, and the adhesive region **224-6**. Vibration actuators (coupling portions) may also be disposed in the non-adhesive region **264** and the non-adhesive region **265**. The adhesive regions **224-1**, **224-2**, **224-3**, and **224-4** may be omitted. Here, four grid-shaped non-adhesive regions are provided, but more grid-shaped non-adhesive regions may be arranged by providing more adhesive regions. Some non-adhesive regions may be not provided with vibration actuators (coupling portions).
(42) [Fourth Modification]
(43) Next, a fourth modification will be described. The fourth modification is in common with the above-described configuration example. Accordingly, different points will be mainly described, and description of common points will be omitted. The same reference numerals are used for components having the same configurations as those of the above-described configuration example.
(44) FIG. **10** is a back view of the transmission panel **230**, the adhesive layer **220**, the first coupling portion **241**, and the second coupling portion **242** of the display speaker **200** according to the fourth modification. Similar to the above configuration example, the display speaker **200** of the fourth modification includes the adhesive layer **220** including the first adhesive region **221**, the second adhesive region **222**, and the third adhesive region **223**. The width of the second adhesive region **222** in the x direction is larger than the width of the first adhesive region **221** and the third adhesive region **223** in the x direction. For example, the width of the second adhesive region **222** in the x direction is twice the width of the first adhesive region **221** and the third adhesive region **223** in the x direction. Increasing the width of the second adhesive region **222** in the x direction further facilitates separation between the vibration caused by the first vibration actuator **251** and the vibration caused by the second vibration actuator **252**.
(45) [Others]
(46) The non-adhesive regions may not be completely separated by the adhesive regions, and adjacent non-adhesive regions may be connected to each other by a non-adhesive portion provided in the adhesive regions. The non-adhesive portion provided in the adhesive regions does not exceed, for example, half the area of the adhesive regions. The adhesive regions around the display panel **210** (for example, the first adhesive region **221**, the third adhesive region **223**, or the like) may be provided with a non-adhesive portion. Increasing the size of the non-adhesive regions and the non-adhesive portions further facilitates the vibration of the transmission panel **230**.
(47) Here, the display speaker **200** outputs stereophonic sound, but sound related to an image (object) displayed on the display panel **210** may be output in conjunction with the image from the position of the image. That is, the display speaker **200** outputs sound related to the image from the position of the image by inputting a signal of sound related to the image to the vibration actuator present at the position of the image in a plan view.
Operation and Effects of Embodiments
(48) The display speaker **200** of the present embodiment includes the display panel **210**, the adhesive layer **220**, the transmission panel **230**, the first coupling portion **241**, the second coupling

portion **242**, the first vibration actuator **251**, and the second vibration actuator **252**. The back side of the display panel **210** and the front side of the transmission panel **230** are joined to each other via the adhesive layer **220**. The back side of the transmission panel **230** and the front sides of the first vibration actuator **251** and the second vibration actuator **252** are joined to each other via the first coupling portion **241** and the second coupling portion **242**. The adhesive layer **220** includes a plurality of strip-shaped regions. The adhesive regions of the adhesive layer **220** do not overlap with the coupling portions in a plan view. The coupling portions are arranged in the non-adhesive regions in a plan view. Since in a plan view the transmission panel **230** and the display panel **210** are not joined at the positions where the vibration actuators are joined to the transmission panel **230**, resistance to vibration is prevented and the display panel **210** is easily vibrated. Separating the first non-adhesive region **261** and the second non-adhesive region **262** facilitates separation between the vibration caused by the first vibration actuator **251** and the vibration caused by the second vibration actuator **252**. That is, the vibration caused by the first vibration actuator **251** and the vibration caused by the second vibration actuator **252** are less likely to interfere with each other. This improves separation of sound output from the display speaker **200** having the single display panel **210**. According to the display speaker **200**, it is possible to provide a panel speaker that separates the left and right signals to improve the quality of output stereophonic sound. In addition, according to the display speaker **200**, it is not necessary to provide a rib (wall) on the vibration panel to partition the vibration actuators as in the related art.
(49) Although the embodiment of the present invention has been described above, this embodiment is merely an example, the present invention is not limited thereto, and various modifications based on the knowledge of those skilled in the art can be made without departing from the gist of the claims. The above configuration examples and modifications can be implemented in combination as much as possible.
REFERENCE SIGNS LIST
(50) **200** display speaker **210** display panel **220** adhesive layer **221** first adhesive region **222** second adhesive region **230** transmission panel **241** first coupling portion **242** second coupling portion **251** first vibration actuator **252** second vibration actuator **800** display speaker of comparative example **810** display panel **820** adhesive layer **830** transmission panel **841** first coupling portion **842** second coupling portion **851** first vibration actuator **852** second vibration actuator

Claims

1. A panel speaker comprising: a single vibration panel formed by adhering to each other one surface of a first panel and one surface of a second panel; a first vibration actuator configured to drive and vibrate the single vibration panel; and a second vibration actuator configured to drive and vibrate the single vibration panel, wherein the single vibration panel has, in a plan view, a first non-adhesive region and a second non-adhesive region in both of which the first panel and the second panel are not bonded to each other, the first vibration actuator is coupled to another surface of the second panel to drive the first non-adhesive region of the single vibration panel, and the second vibration actuator is coupled to the other surface of the second panel to drive the second non-adhesive region of the single vibration panel, and the first vibration actuator and the second vibration actuator are not directly coupled to the first panel.
2. The panel speaker according to claim 1, wherein the first non-adhesive region and the second non-adhesive region are arranged, in the plan view, at both sides of a virtual straight line passing through a center in a longitudinal direction of the single vibration panel and being orthogonal to the longitudinal direction.
3. The panel speaker according to claim 2, wherein the first non-adhesive region and the second non-adhesive region are arranged, in the plan view, in line symmetry with respect to the virtual straight line.

4. The panel speaker according to claim 3, wherein the first vibration actuator is configured to receive a left sound signal of stereophonic sound, and the second vibration actuator is configured to receive a right sound signal of stereophonic sound.
5. The panel speaker according to claim 3, wherein the single vibration panel includes a third non-adhesive region that, in the plan view, separates the first non-adhesive region and the second non-adhesive region from each other.
6. The panel speaker according to claim 3, wherein the single vibration panel has a plurality of non-adhesive regions in each of which the first panel and the second panel are not bonded to each other, and the plurality of non-adhesive regions are, in the plan view, partitioned in a lattice shape and include the first non-adhesive region and the second non-adhesive region.
7. The panel speaker according to claim 3, wherein the first panel is a display panel.
8. The panel speaker according to claim 7, wherein the first panel is an organic light emitting diode panel.
9. The panel speaker according to claim 2, wherein the first vibration actuator is configured to receive a left sound signal of stereophonic sound, and the second vibration actuator is configured to receive a right sound signal of stereophonic sound.
10. The panel speaker according to claim 2, wherein the single vibration panel includes a third non-adhesive region that, in the plan view, separates the first non-adhesive region and the second non-adhesive region from each other.
11. The panel speaker according to claim 2, wherein the single vibration panel has a plurality of non-adhesive regions in each of which the first panel and the second panel are not bonded to each other, and the plurality of non-adhesive regions are, in the plan view, partitioned in a lattice shape and include the first non-adhesive region and the second non-adhesive region.
12. The panel speaker according to claim 2, wherein the first panel is a display panel.
13. The panel speaker according to claim 12, wherein the first panel is an organic light emitting diode panel.
14. The panel speaker according to claim 1, wherein the first vibration actuator is configured to receive a left sound signal of stereophonic sound, and the second vibration actuator is configured to receive a right sound signal of stereophonic sound.
15. The panel speaker according to claim 1, wherein the single vibration panel includes a third non-adhesive region that, in the plan view, separates the first non-adhesive region and the second non-adhesive region from each other.
16. The panel speaker according to claim 1, wherein the single vibration panel has a plurality of non-adhesive regions in each of which the first panel and the second panel are not bonded to each other, and the plurality of non-adhesive regions are, in the plan view, partitioned in a lattice shape and include the first non-adhesive region and the second non-adhesive region.
17. The panel speaker according to claim 1, wherein the first panel is a display panel.
18. The panel speaker according to claim 17, wherein the first panel is an organic light emitting diode panel.
19. A display panel speaker for a-stereo sound, comprising: a display panel; a vibration propagation panel; stripe shaped first, second and third adhesive patterns formed between the display panel and the vibration propagation panel; a first vibration actuator that (i) is disposed between the first and the second adhesive patterns, (ii) is adhered to the vibration propagation panel to drive and vibrate the vibration propagation panel, and (iii) is not directly adhered to the display panel; and a second vibration actuator that (i) is disposed between the second and the third adhesive patterns, (ii) is adhered to the vibration propagation panel to drive and vibrate the vibration propagation panel, and (iii) is not directly adhered to the display panel.